

Supporting Information

FEPrepare: A Web-based Tool for Automating the Setup of Relative Binding Free Energy Calculations

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FREE ENERGY PERTURBATION (FEP)

The free energy of binding of a ligand L to a protein P is given by:



The binding constant K_B° is provided by the ratio of concentrations of product and reactants of equation 1:

$$K_B^\circ = C_0 \frac{[PL]}{[P][L]}, \quad (2)$$

where C_0 is the standard state concentration (usually 1M). The standard free energy of binding ΔG_{bind}° can be computed by a statistical thermodynamics approach through the following equation¹:

$$\Delta G_{bind}^\circ = -k_B T \ln K_B^\circ = \frac{Z_{PL} Z_{solv} V}{Z_P Z_L V_0}, \quad (3)$$

where k_B is the Boltzmann constant and T is the temperature. Z_{PL} , Z_P , Z_L , and Z_{solv} represent the configuration integrals of the protein-ligand complex, protein, ligand, and solvent, respectively. Finally, V is the volume available for the bound ligand and V_0 the standard volume (usually 1 dm³ mol⁻¹). Thus, the computation of the standard binding free energy depends on solving the different configuration integrals of the system, for instance the configurational integral of the protein-ligand complex is given by the following equation:

$$Z_{PL} = \int \exp(-\beta U_{PL}(q)) dq, \quad (4)$$

where $U_{PL}(q)$ is the potential energy of conformation q of the protein-ligand complex. However, direct computation of these integrals is numerically impossible, due to the high

dimensionality of the protein-ligand complexes. Instead, it is much easier to compute ratios of configurational integrals between two structurally related ligands A and B.

FEPprepare NAMD/FEP WORKFLOW

Hybrid RTF & PDB creation, and protein-ligand complex files.

The atoms with the new charges will be included in the hybrid RTF with different names from their reference counterparts, in order for NAMD to recognise them as different atoms. The produced hybrid RTF will include the common part of the two ligands, the perturbation area before the transformation, and the perturbation area after the transformation. So, essentially, it will include the whole reference ligand and, from the perturbed ligand, only the perturbed atoms that were affected by the partial charge distribution procedure. In addition to a hybrid RTF, a hybrid PDB file needs to be produced, accordingly.

Because the perturbed atoms that had their charges changed had also their names changed, with respect to their reference counterparts, the PRM file of the perturbed ligand needs to be updated, to account for the new names.

For the complex leg of the simulation, it is necessary to have a protein-ligand complex PDB file. To build the complex PDB, the protein PDB and the hybrid PDB files need to be combined, simply by concatenating the two files and fixing the atom numbering of the ligand at the end of the new file.

STATISTICS CONCERNING THE DIV CUTOFF

An important part of the FEPrepare algorithm concerns the charge spreading scheme. The DIV cutoff described in the main text, is used to define the charge that will be dispensed among the perturbed atoms. This DIV has been calibrated to be valid for maintaining the same charge within the perturbation and it acts as a cutoff threshold for the convergence of the FEP algorithm. For this purpose, we set this threshold to 0.02.

One main concern of this approach is the magnitude of the charge that end up getting spread. For this purpose, a set of 20 perturbations was examined using the 0.02 cutoff in order to define the number of atoms that change in order to achieve this threshold. The examined molecules are depicted in Figure S4 and the number of changing atoms per perturbations is shown in Table S1.

Table S1. Number of changing atoms per perturbation.

Perturbation	Number of atoms changing
CK666→AI003	3
AI003→CK666	4
CDK 17→18	6
CDK 18→17	7
CDK 19→20	4

CDK 24→25	4
CDK 34→36	3
CDK 34→35	6
CDK 37→38	3
CDK 38→37	3
CDK 39→40	4
CDK 42→43	2
CDK 43→42	3
HIF2alpha 43→338	9
Example1→Example2	6
Example2→Example1	8
Example1→Example3	7
Example3→Example1	7
Example2→Example3	3
Example3→Example2	5

TEST CASES

Our tests were conducted on two model systems in order to better understand the web-server outputs along with their validation. We used these outputs to perform FEP/MD calculations using the dual topology protocol of NAMD. We then compared the results of these calculations with relative binding free energies obtained from the FEP+ module in the Schrodinger Suite² and from experimental data.

Arp2/3

Actin-related protein 2/3 (Arp2/3) complex is a seven-subunit assembly that plays an important role in the regulation of the actin skeleton. It serves as an actin nucleator in eukaryotes as it has the capability of organizing filaments into branched networks. This is important for the generation of actin networks and is crucial for processes-like cell motility, phagocytosis, and intracellular motility of lipid vesicles.³

The inhibition of Arp2/3 has shown to lead to the control of plasticity of nerve synapses,⁴ contribute to the predominance of the helper with regard to lamellipodiums,⁵ regulate the shape and movement of the endosomes,⁶ and change the mechanism of regulation of ion channels by the cortactin protein.⁷

A known inhibitor for Arp2/3 is CK666 (Figure S6) and was tested against the perturbed ligand AI003 (Figure S7). We have also performed the backward calculation for the reference (CK666) and perturbed (AI003) ligand. The server provided similar topologies with the forward calculation. Finally, the backward and forward calculations are within the error of the method providing the same result with opposite signs, as seen in Table S1.

CDK8

The Cyclin-dependent kinase 8 (CDK8) is an enzyme that belongs to the CDK family and is encoded by the *CDK8* gene.^{8,9} It associates with the mediator complex and acts as a transcription-regulating kinase. CDK8 regulates the WNT/ β -Catenin signalling pathway through activation of β -catenin that drives colon tumorigenesis.¹⁰ Thus, CDK8 functions as an oncogene in colon cancer cells.

Inhibition of CDK8 has been suggested in order to treat diseases associated with the WNT/ β -Catenin signalling pathway.^{11,12} Potent and selective CDK8 inhibitors have been discovered and optimised by Schiemann et al.¹³ and they have been included to the Merck KGaA dataset. This is a benchmark dataset of pharmaceutically relevant targets for relative free energy calculations created by Christina Schindler and Daniel Kuhn.^{14,15}

The perturbation between ligand 34 (Figure S8) and 36 (Figure S9) was examined using the FEPPrepare server.

As explained in main text, after the user has uploaded the required files to FEPPrepare, the algorithm downloads the files, and parses them. All atoms in both ligands are the same, apart from atoms **F00** and **F01** (in ligand 34), and **H0A**, **O0B**, **C0C**, **H0F**, **H0D** and **H0E** (in ligand 36). This is the area that takes part in the perturbation.

Besides the small modification in the atoms, an extra charge is introduced in the area. This time we have to include one more atom to our calculation. This atom, **C23**, is around the area that takes part in the perturbation; therefore, its charge changes significantly during the transformation. The manual calculations agree that the output is correct.

Now the hybrid.rtf and hybrid.pdb files have been created and the PRM file of ligand 36 has been updated. So, the next step is to combine the protein PDB file with the hybrid.pdb file in order to create the complex.pdb file. The code generates correctly the files that VMD requires to run its calculations and to create the PSF files.

Results

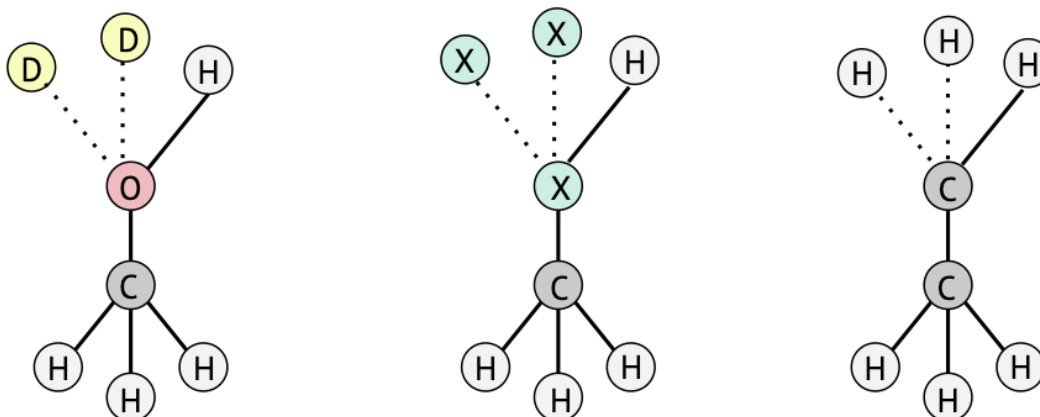
The outputs obtained by FEPrepare for all perturbations are totally compatible with the NAMD software since the simulations run and finish correctly. The results obtained from the NAMD software, the relative binding free energies from FEP+, and the experimental measurements for both perturbations are illustrated in the following table:

Table S2. Results for the relative binding free energies of two protein-ligand complexes obtained from the NAMD and FEP+ software compared to experimental measurements. Energies and standard deviations (std) are reported in kcal/mol. The forward and backward calculation of CK666-AI003 is also reported.

	$\Delta\Delta G_{\text{NAMD}}$	std	$\Delta\Delta G_{\text{FEP+}}$	std	$\Delta\Delta G_{\text{experimental}}$
CK666-AI003 (forward)	-4.07	0.06	-4.50	0.41	-1.17
AI003-CK666 (backward)	3.61	0.06			-1.17
CDK 34-36	1.81	0.11	-0.47	0.20	-0.09

SUPPLEMENTARY FIGURES

a) Single Topology



b) Dual Topology

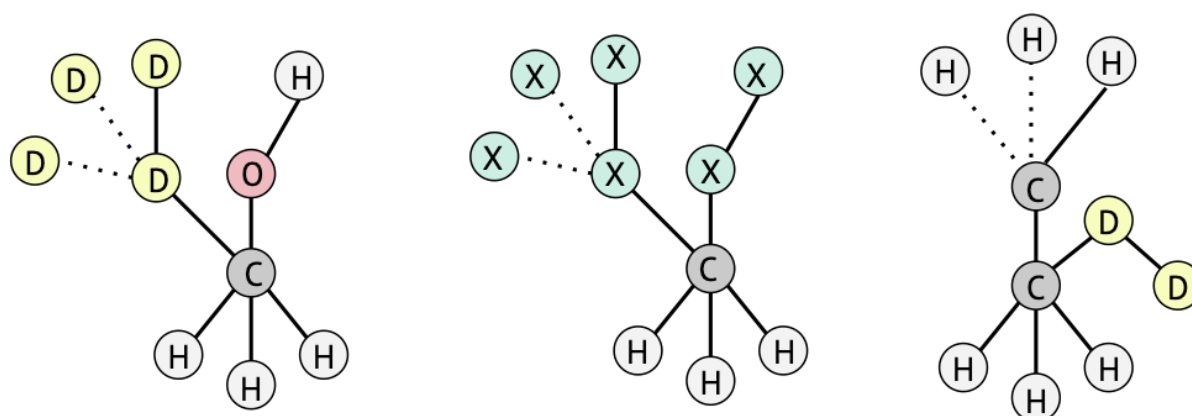


Figure S1. Single-Dual Topology approach for the methanol-ethane perturbations. X represents the alchemical intermediate atoms, while D represents the noninteracting dummy atoms. In the single topology scheme (a, upper), at the one end-state (on the left), the upper heavy atom is an oxygen, and two hydrogen atoms are transformed to noninteracting dummy atoms. At the other end-state (on the right), the upper heavy atom is a carbon, and all atoms are fully interacting. On the contrary, in the dual topology approach (b, lower), no atoms change type. At the methanol end-state, the methyl group has its interactions turned off, whereas in the ethane end-state, the hydroxy group is noninteracting.

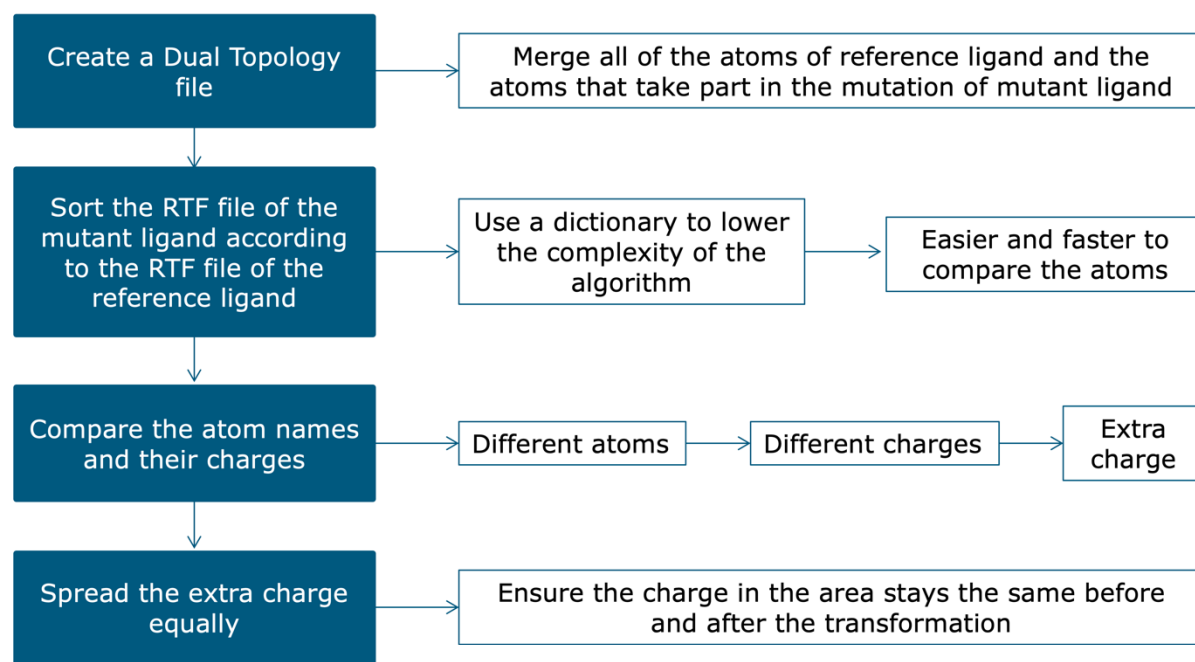


Figure S2. Detailed description of the Dual Topology workflow used for the creation of the hybrid files.

Spreading the extra charge

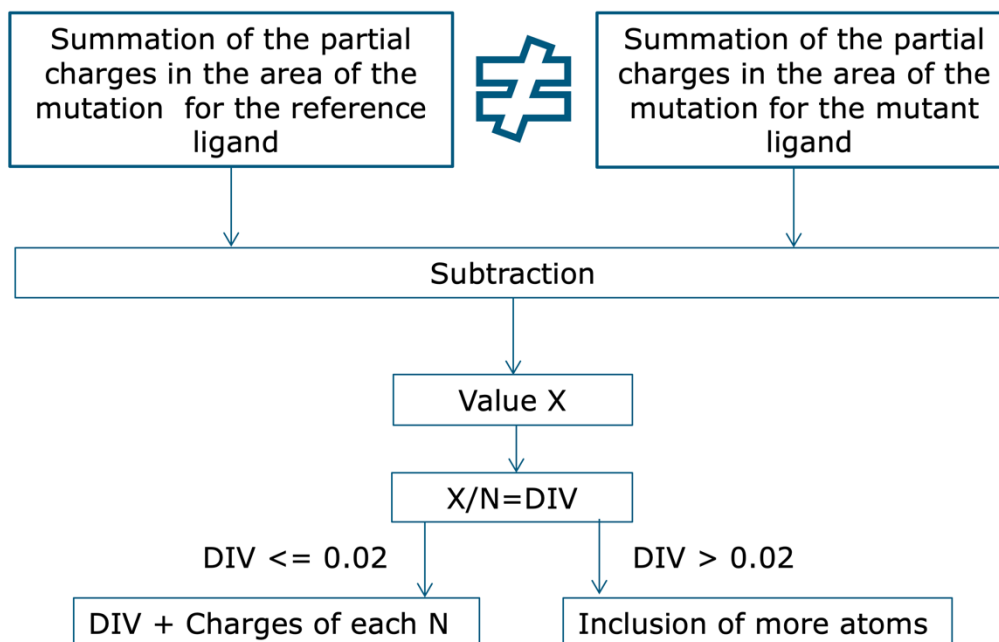


Figure S3. Partial Charge re-distribution scheme. The first step is to sum up the partial charges of the atoms of the reference (Sum A) and the perturbed ligand (Sum B) that take part in the perturbation. Then these values are subtracted in order to obtain the extra charge X that is going to be distributed among the atoms of the perturbed ligand. Afterwards X is divided by the number of atoms that take part in the perturbation in order to estimate the cutoff threshold for the convergence of the FEP algorithm. If this number is higher than 0.02, then new neighbouring-to-the- perturbation atoms are included in the calculation. This procedure continues recursively until DIV is lower than 0.02.

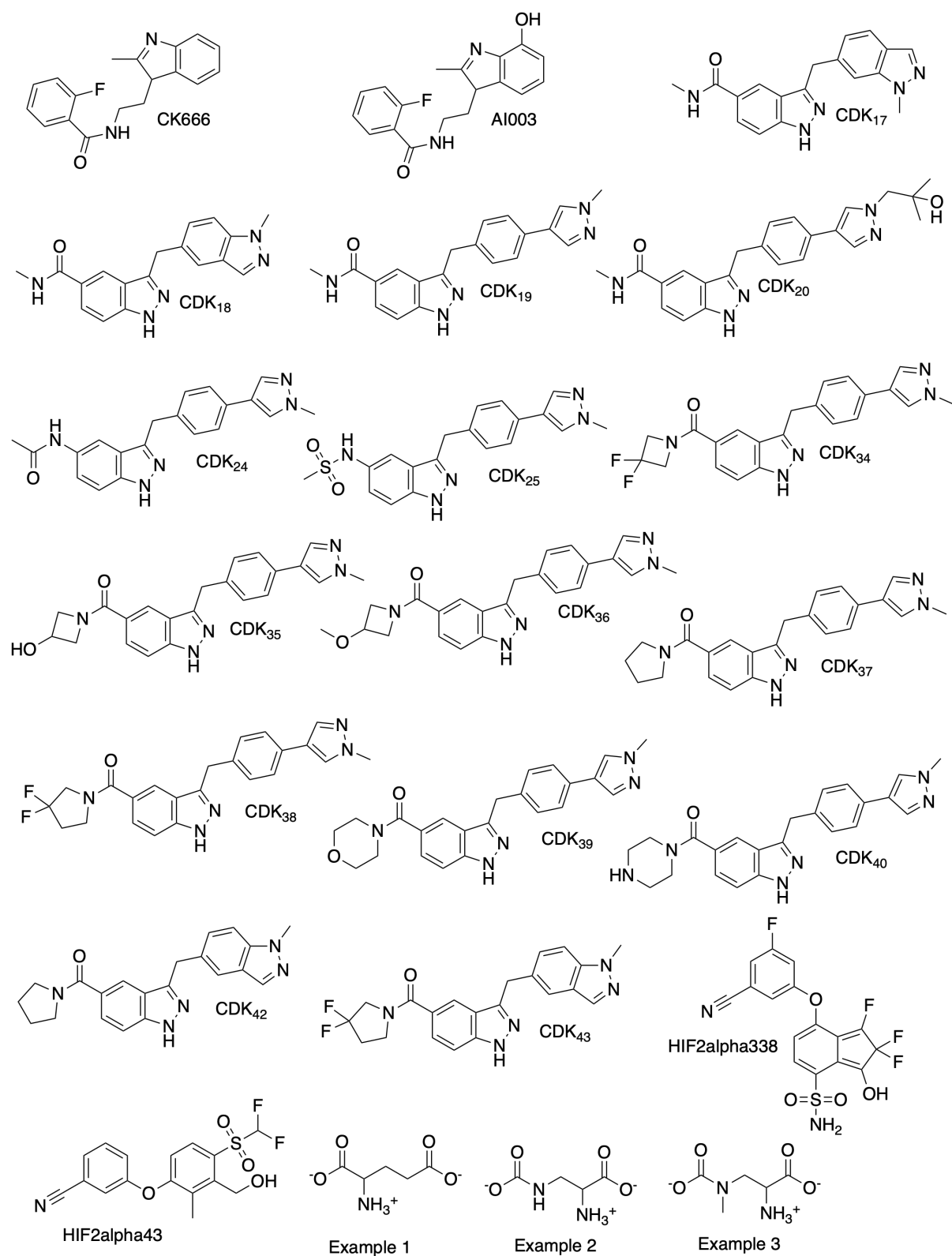


Figure S4. 2D structures of the ligands taking part in the examined perturbations.

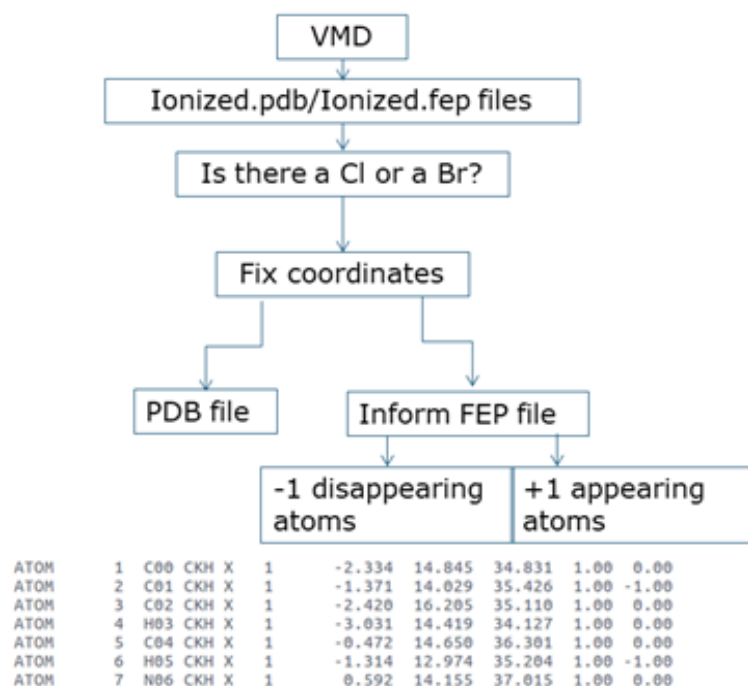


Figure S5. Workflow for the generation of the final PDB and FEP files for the FEP/MD calculations. The script searches the generated PDB and FEP files from VMD for Cl or Br atoms. If such an atom is found, it reads the correct coordinates from the initial PDB files and updates the final ionized.pdb and ionized.fep files. Then, the algorithm modifies the B-factor column of the ionized.fep file by setting the values of the disappearing atoms to -1.00 and the values of the appearing atoms to 1.00. The final PSF, PDB, and FEP files needed from NAMD to perform FEP/MD calculations are ionized.psf, ionized_solvent.pdb, and ionized_solvent.fep, respectively, for the free phase, and ionized.psf, ionized_complex.pdb, and ionized_complex.fep, respectively, for the bound phase.

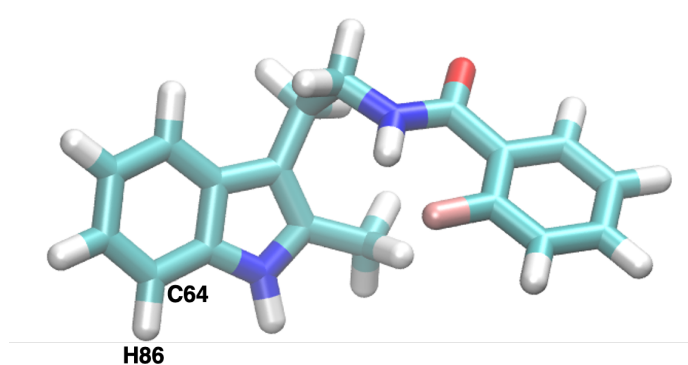


Figure S6. 3D structure of the CK666 inhibitor of the Arp2/3 complex protein. Labelled atoms C64 and H86 illustrate the area that will take part in the CK666-AI003 perturbation.

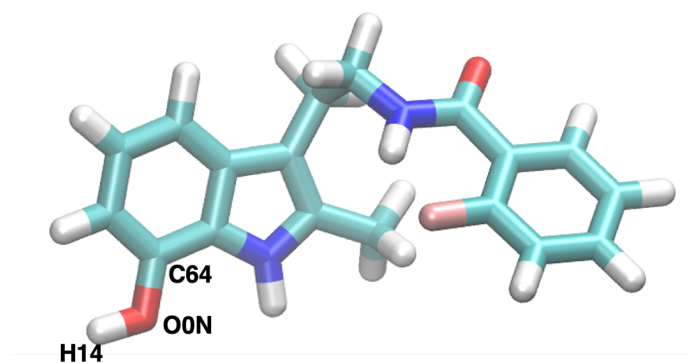


Figure S7. 3D structure of the AI003 inhibitor of the Arp2/3 complex protein. The atoms labelled as C64, O0N, and H14 are the perturbed atoms during the CK666-AI003 perturbation.

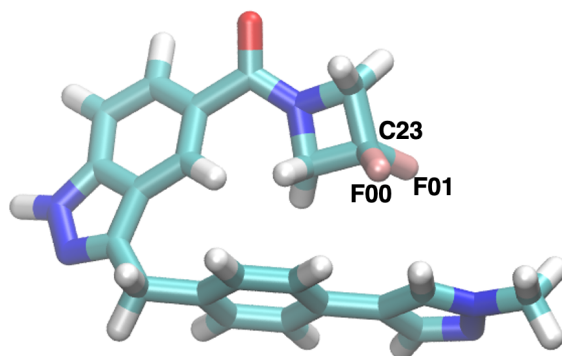


Figure S8. 3D structure of ligand 34 of the CDK8 protein. Labelled atoms C23, F00, and F01 illustrate the area that will take part in the 34-36 perturbation.

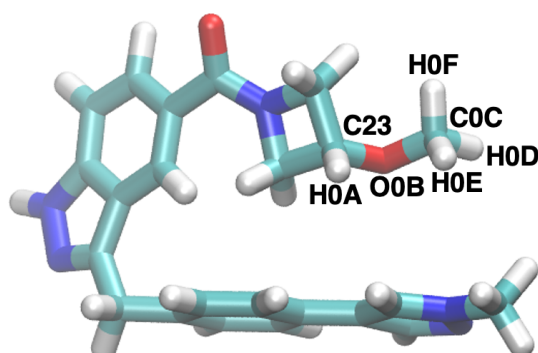
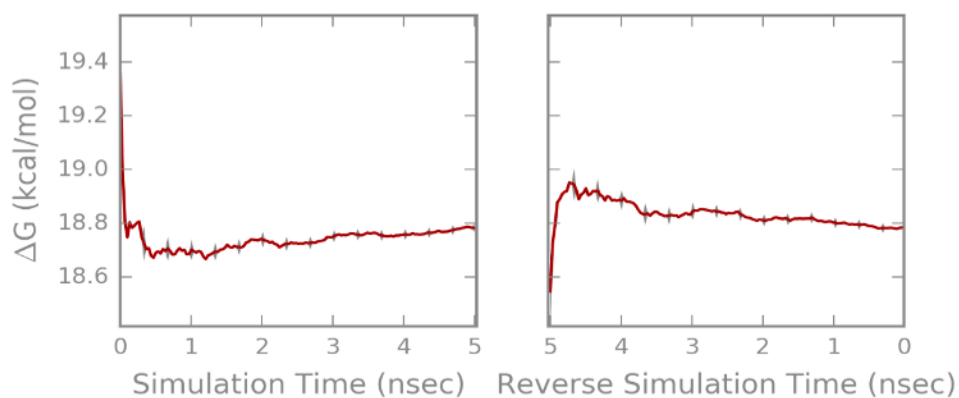


Figure S9. 3D structure of ligand 36 of the CDK8 protein. The atoms labelled as C23, H0A, O0B, C0C, H0E, H0F, and H0D are the perturbed atoms during the 34-36 perturbation.

Free Energy Convergence

Solvent Leg



Complex Leg

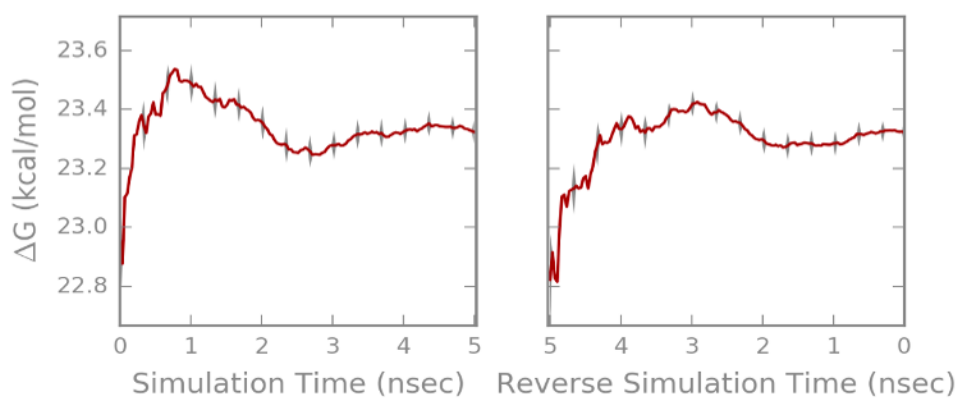


Figure S10. Convergence plots for the calculated binding free energies in solvent and complex legs of the CK666-AI003 perturbation obtained from FEP+ as a function of time. Two plots for each leg demonstrate the cumulative data during different time window schemes: forward and reverse.

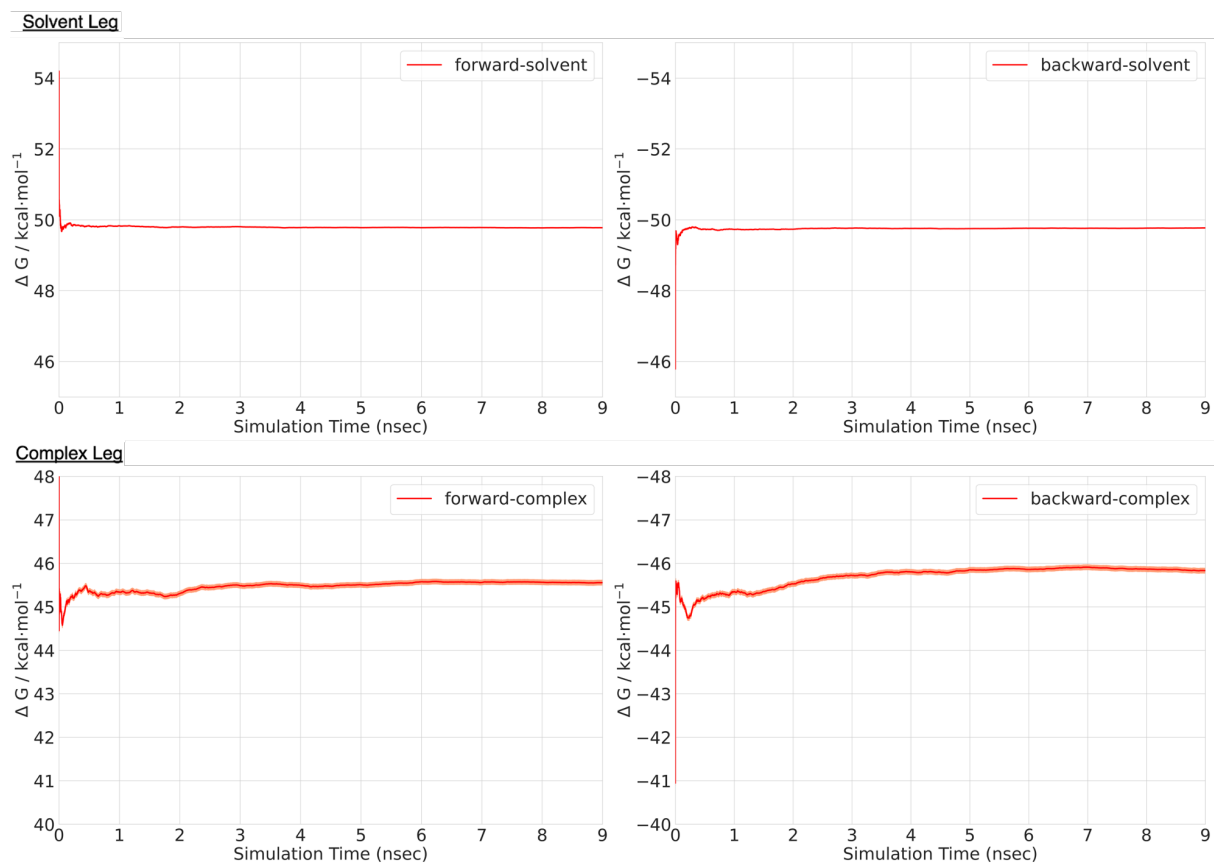


Figure S11. Convergence plots for the calculated binding free energies in solvent and complex legs of the CK666-AI003 perturbation obtained from NAMD are plotted as a function of time. Two plots for each leg demonstrate the cumulative data during different time window schemes: forward and reverse.

THE README FILE

FEPPrepare README file

Please cite this as Zavitsanou et al., JCIM, 2021

This folder should contain:

- Two subfolders, “complex” and “solvent”.
- Six renamed files (PDB, RTF, PRM) for the parent and the mutated ligand (reference.pdb, reference.rtf, reference.prm, mutant.pdb, mutant.rtf, mutant.prm).
- Two hybrid files for both parent and mutated ligand following the dual topology implementation of NAMD, named as “ligand.pdb” and “ligand.rtf”.
- The renamed PRM file after the creation of the two hybrid files, “updated.prm”.
- The “complex.pdb” file.
- The “fep.tcl” script that NAMD needs to run the simulation.
- A file with the OPLS-AA parameters of proteins as downloaded from <http://zarbi.chem.yale.edu/oplsaam.html> (par_opls_aam.inp).
- A file with the OPLS-AA topology of proteins as downloaded from <http://zarbi.chem.yale.edu/oplsaam.html> (top_opls_aam.inp).

The “complex” folder contains the following files:

- The PDB files for each of the protein chains and the ligand (in the case of the example “chainA.pdb” and “chainB.pdb” for the protein and “chainX.pdb” for the ligand).
- The files required to run VMD, “psfgen”, and “VMD_prepare_complex_after_gui_autopsf” or “VMD_prepare_complex_after_gui_autopsf_ionized” in case the user requested 150mM NaCl salt addition.
- The files that VMD provides as output, i.e. “complex_wb.log”, “complex_wb.pdb”, “complex_wb.psf”, “psf-complex.psf”, “psf-complex.pdb”, which are required for VMD to generate the final files for the FEP calculation.
- Files named “ionized.pdb”, “ionized.fep”, “ionized.psf” as generated by VMD.
- Files named “ionized_complex.pdb” and “ionized_complex.fep”, which contain the updated ionized files, after implementing the FEPPrepare NAMD/FEP Workflow (see main text for details).
- Two text files: a) “min-max_center”, which provides the cell basis vectors and the cell origin to be included in the NVT equilibration input file, and b) “vmd_log”, which describes all the steps performed by VMD.
- A file named “complex-input-files.tar.gz” which provides you with the configuration files needed to run NAMD.

The “solvent” folder contains the following files:

- The files used to run VMD, “psfgen_solv”, and “VMD_prepare_ligand_after_gui_autopsf” or “VMD_prepare_ligand_after_gui_autopsf_ionized” in case the user requested 150mM NaCl salt addition.
- The files that VMD gives as an output, “ligand_wb.log”, “ligand_wb.pdb”, “ligand_wb.psf”, “psf-solvated.psf”, “psf-solvated.pdb”, which are required for VMD to generate the final files for the FEP calculation.
- Files named “ionized.pdb”, “ionized.fep”, “ionized.psf” as generated by VMD.

- Files named “ionized_solvent.pdb” and “ionized_solvent.fep” which contain the updated ionized files, after implementing the FEPrepare NAMD/FEP Workflow (see main text for details).
- Two text files: a) “min-max_center”, which provides the cell basis vectors and the cell origin to be included in the NVT equilibration input file, and b) “vmd_log”, which describes all the steps performed by VMD.
- A file named “solvent-input-files.tar.gz” which provides you with the configuration files needed to run NAMD

These files are provided to the user so that they have full control of the FEPrepare workflow and VMD-generated files.

The actual files needed for the NAMD/FEP calculation are “ionized_complex.pdb”, “ionized_complex.fep”, “ionized_solvent.pdb”, “ionized_solvent.fep”, “ionized.psf”.

Your data will be deleted after two days. No other user has access to your data.

In case you do not see all the files mentioned above, there was probably an error during the calculation. Please check that you have provided the correct input or contact us at zavitsanoustamatia@gmail.com or zcournia@bioacademy.gr.

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